

Unit 3 — Numerical Methods (4 Qs)

1. RK-4 Method — solve $dy/dx = x+y$ (or similar), $y(0)=1$, find $y(0.1)/y(0.2)$, $h=0.1$ — P $\approx$ 95%

Also commonly asked as: $dy/dx = (y^2 - x^2) / (y^2 + x^2)$, $y(0)=1$, $h=0.2$

Scoring Breakdown:

- State x_0 , y_0 , h , $f(x,y) \rightarrow$ compute k_1 (1 mark)
- Compute k_2 , k_3 (2.5 marks)
- Compute k_4 and final y_{n+1} (3.5 marks)

2. Milne's Predictor-Corrector Method — P $\approx$ 95%

Classic form: $dy/dx = x - y^2$, given $y(0)$, $y(0.1)$, $y(0.2)$, $y(0.3)$, find $y(0.4)$

Also seen as: $5xy + y^2 - 2 = 0$, $y(4)=1$, ... find $y(4.4)$ (exact PYQ + QB match)

Scoring Breakdown:

- Tabulate x_n , y_n , f_n
- Predictor formula (label clearly) $\rightarrow$ predicted slope $f_4^{(P)}$
- Corrector formula $\rightarrow$ final corrected value

3. Euler's Modified Method (iterative convergence) — P $\approx$ 90%

Problem: $dy/dx = x+y$, $y(0)=1$, find $y(0.2)$, $h=0.1$ (or $dy/dx = 1-y / x+\sin y$ variants)

Scoring Breakdown:

- Predictor $y_1^{(0)}$
- Iterate corrector until 2 consecutive values agree to 4 decimals
- Repeat same process for step 2 (to find $y(0.2)$)

4. Taylor's Series Method with exact-solution comparison — P ≈ 85%

Problem: $dy/dx = 2y + 3e^x$, $y(0)=0$, find $y(0.2)$, compare with exact solution

Scoring Breakdown:

- Differentiate successively for $y'(0)$, $y''(0)$, $y'''(0)$, $y^{(4)}(0)$
- Substitute in Taylor expansion
- Solve exact linear ODE (IF = e^{-2x})
- Compare, show error is negligible

Unit 4 — Laplace & Fourier Transforms (5 Qs)**5. Convolution Theorem — state, prove, and evaluate $L^{-1}\{s / (s^2+a^2)^2\}$ — P ≈ 95%**

Scoring Breakdown:

- State theorem (1 mark)
- Prove via double integral / change of variables (3 marks)
- Set $F(s) = s/(s^2+a^2) \rightarrow f(t) = \cos(at)$, $G(s) = 1/(s^2+a^2) \rightarrow g(t) = (1/a)\sin(at)$
- Convolution integral with trig identity $2\cos A \sin B \rightarrow$ evaluate to $(t/2a)\sin(at)$ (3 marks)

6. Solve $(D^2+9)y = \cos(2t)$, $y(0)=1$, $y(\pi/2)=-1$ via Laplace — P ≈ 90%

Scoring Breakdown:

- Take LT: $[s^2Y(s) - sy(0) - y'(0)] + 9Y(s) = s/(s^2+4)$
- Substitute $y(0)=1$, let $y'(0)=c$
- Partial fractions $\rightarrow$ invert
- Use $t=\pi/2$, $y=-1$ to solve for c ($= 12/5$)

7. $L^{-1}\{s / [(s^2+1)(s^2+4)]\}$ via convolution — P ≈ 85%

Notes: Same convolution-theorem family as Q5, but simpler denominator.

Good backup/alternate question.

8. Evaluate $\int_0^\infty (\cos(at) - \cos(bt))/t \, dt$ OR $\int_0^\infty e^{-t} \sin^2(t)/t \, dt$ (division-by-t property) — P $\approx$ 85%

Scoring Breakdown:

- Write $f(t) = \sin^2(t) = (1 - \cos(2t))/2$, find $F(s)$
- Apply $L\{f(t)/t\} = \int_s^\infty F(u) \, du$
- Integrate/simplify to log form
- Substitute $s=1$ for final numeric answer (for the e^{-t} case)

9. Fourier Sine Transform of e^{-ax}/x ; deduce $\int_0^\infty (e^{-ax} \sin(sx))/x \, dx = \tan^{-1}(s/a)$ — P $\approx$ 80%

Scoring Breakdown:

- Set up $I(s)$, differentiate under integral sign (Leibniz)
- Standard exponential-cosine formula
- Integrate w.r.t. s
- Evaluate constant at $s=0$

Unit 5 — Probability & Statistics (4 Qs)**10. Fit Poisson Distribution: r: 0-4, f: 122, 60, 15, 2, 1 ($e^{-0.5} = 0.6065$) — P $\approx$ 95%**

Scoring Breakdown:

- Create Table $(x, f, fx) \rightarrow N = 200, \Sigma fx = 100$
- Mean $\lambda = 100/200 = 0.5$
- Calculate $T(0) = 200 \times e^{-0.5} \approx 121$
- Use recurrence relation $T(x+1) = T(x) \times [\lambda / (x+1)]$ for remaining values

11. Derive Mean and Variance of Poisson Distribution ($= \lambda$ both) — P $\approx$ 85%

Scoring Breakdown:

- Mean via $E[X] = \Sigma x \cdot P(x)$
- Cancel x with $x!$, recognize e^λ series $\rightarrow \lambda$
- Variance via $E[X(X-1)] + E[X] - (E[X])^2$
- $\lambda^2 + \lambda - \lambda^2 = \lambda$

12. Given $f(x) = cx^2$, $0 < x < 1$ — find c , and $P(1/3 < X < 1/2)$ — $P \approx 85\%$

Scoring Breakdown:

- $\int_0^1 cx^2 dx = 1 \rightarrow c = 3$
- $P = \int_{1/3}^{1/2} 3x^2 dx = [x^3]$
- Careful fraction arithmetic $\rightarrow (1/8) - (1/27) = 19/216 \approx 0.088$

13. Normal Distribution: mean = 65.5", SD = 6.2", find % between 54.8" and 68.8" — $P \approx 85\%$

Scoring Breakdown:

- Use $Z = (X - \mu) / \sigma \rightarrow Z_1 = -1.73, Z_2 = +0.53$
- Split area across mean $\rightarrow$ sum table values $(0.4582 + 0.2019)$
- Total area = 0.6601 $\rightarrow 66.01\%$